

SDST3612 — Final Project

Brain Tumor Classification

Poon Tsz Chung
Chan Ronny Ryan
Liao Wang
Ko Cheung Tsun
Myriam Akrouit

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1 Introduction

The objective of this project is to develop statistical machine learning models for presurgical brain tumor classification using multimodal data. The task is a five-class classification problem involving Brain Metastase Tumour, Glioma, Meningioma, Pineal tumour and Choroid plexus tumour, and Tumors of the sellar region. The dataset contains 2,838 patients and is divided into 1,983 training cases, 283 validation cases, and 572 test cases. The available modalities include deep image features extracted from MRI, radiomics features, clinical and demographic variables, and radiology reports. The original MRI volumes are not released; instead, representative tumor-containing 2D slices were used to derive features.

In medical imaging, brain tumor differentiation is challenging because several tumor types share subtle morphological characteristics, while some classes are rare and therefore difficult to learn robustly. For this reason, our work did not focus only on building a single predictive model, but also on understanding the contribution and limitations of each modality.

In our group, each teammate first explored one modality independently before moving to multimodal integration. The goal of this first stage was to quantify how much useful information each data source provides on its own, and to identify which modalities are likely to be most complementary in a final fusion model.

Regarding evaluation, although overall system performance is important, class imbalance makes it necessary to go beyond a single summary score. We therefore report not only Micro F1-score, but also Macro F1-score, class-wise precision and recall, and confusion matrices.

2 EDA

2.1 Feature Construction from MRI Image Features

The `image_features` modality consists of deep embeddings extracted from MRI using a pretrained ResNet model. For each patient, up to four MRI modalities are available:

- `ax_t1`,
- `ax_t1c`,
- `ax_t2`,
- `ax_t2f`.

Each available modality is stored as a file `image.npy` and corresponds to a vector of shape `(2048,)` with `float32` values. In other words, the raw input for this modality is not an image anymore, but a compact deep representation extracted from an image. Because of this, the task becomes a high-dimensional tabular classification problem rather than a pixel-level vision problem.

To build one fixed-length representation per patient, we concatenated the four modality vectors. Since not all modalities are available for every patient, we replaced missing modalities by a zero vector of length 2048 and appended four binary modality-presence indicators. This produced one feature vector of dimension

$$4 \times 2048 + 4 = 8196$$

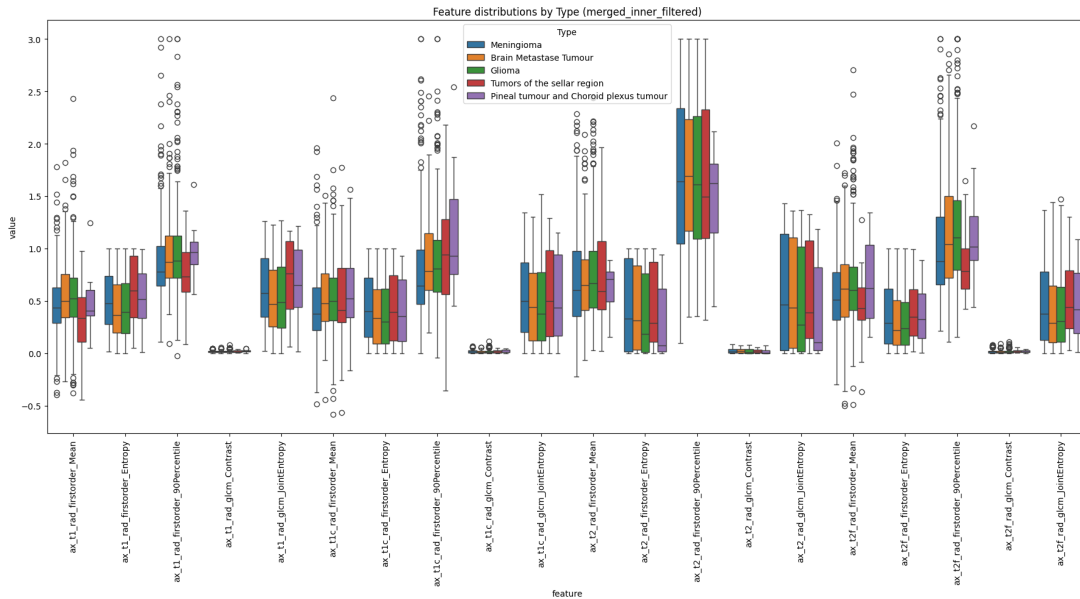
for each patient.

2.2 Feature Construction from MRI Radiomics

The `radiomics_info` extracts extra information from raw `image_features`. For each slices of the `image_features`, 5 features are extracted:

- Mean,
- Entropy,
- 90Percentile,
- Contrast.
- JointEntropy

Visualization :



2.3 Data Distribution and Modality Availability Analysis

Before training the models, we performed a simple exploratory analysis of the modality availability and the class distribution.

First, we verified that the MRI deep feature files were consistent: each available modality was represented by a 2048-dimensional `float32` vector. Then, we counted the number of patients for which each modality was present. On the training set, `ax_t1` was available for all 1,983 cases, while `ax_t1c`, `ax_t2`, and `ax_t2f` were available for 1,899, 1,871, and 1,789 cases respectively. On the validation set, the corresponding counts were 283, 270, 269, and 249. This shows that missing modalities are not negligible and must be handled explicitly in the feature construction pipeline.

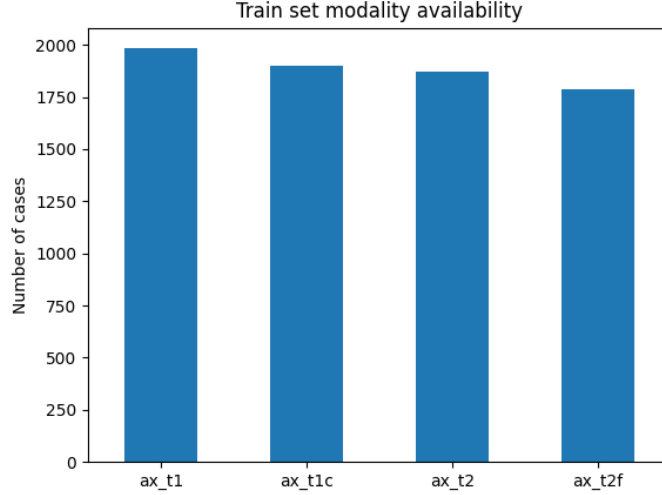


Figure 1: Availability of the four MRI modalities in the training set. T1 is complete, while T1c, T2 and T2-FLAIR are missing for a subset of patients.

Second, we inspected the class distribution. The dataset is strongly imbalanced. In the training set, Glioma and Meningioma dominate with 924 and 728 samples respectively, whereas Tumors of the sellar region and Pineal tumour and Choroid plexus tumour have only 56 and 23 cases. The same imbalance is visible in the validation set.

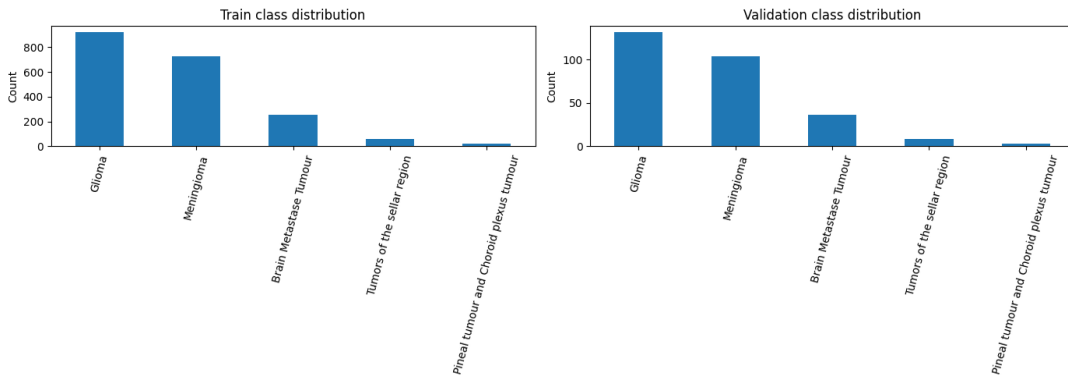


Figure 2: Class distribution in the training and validation sets

3 Separate Training

As a preliminary step toward building combined models, our team trained separate classifiers using each group of features independently: baseline `image_feature`, `radiomics`, and `clinical_information`. In the next step, we aim to combine these models using techniques such as soft voting or naive bayes.

3.1 Baseline Models

To assess the predictive power of the deep image features alone, we trained three baseline classifiers:

- Logistic Regression with feature standardization and balanced class weights,
- Linear Support Vector Classifier (Linear SVC) with feature standardization and balanced class weights,
- Random Forest with balanced subsampling,

The choice of these models was motivated by the structure of the data. Since the MRI inputs are already transformed into high-level 2048-dimensional embeddings by ResNet, linear models are a natural first baseline. At the same time, tree-based ensembles provide a useful comparison to test whether non-linear feature interactions can further improve performance.

3.2 Validation Results

The resulting feature matrices had dimensions (1983,8196) for training and (283,8196) for validation. The validation results are summarized in Table 1.

Model	Macro F1	Weighted F1	Micro F1
Linear SVC (balanced)	0.4617	0.5932	0.5972
Logistic Regression (balanced)	0.4552	0.5959	0.5972
Random Forest (balanced)	0.2405	0.4946	0.5548

Table 1: Validation performance of the baseline models trained only on `image_features`.

The best model was the balanced Linear SVC, with a Micro F1-core of 0.5972 and Macro F1-score of 0.4617. Logistic Regression performed very similarly, while the tree-based model was clearly worse. This suggests that the ResNet embeddings are already informative and may be more effectively separated by linear decision boundaries than by tree ensembles in this high-dimensional setting.

3.3 Class-wise Analysis

To better understand the behavior of the best-performing model, we examined the class-wise precision, recall, and F1-score of the balanced Linear SVC.

Class	Precision	Recall	F1-score
Brain Metastase Tumour	0.3000	0.3333	0.3158
Glioma	0.6393	0.5909	0.6142
Meningioma	0.6581	0.7404	0.6968
Pineal tumour and Choroid plexus tumour	1.0000	0.3333	0.5000
Tumors of the sellar region	0.3333	0.1250	0.1818

Table 2: Class-wise performance of the best unimodal image-features model (Linear SVC).

The results show that the model performs best on Meningioma and Glioma, which are also the most represented classes. In contrast, performance is much weaker for Brain Metastase Tumour and especially for Tumors of the sellar region. Although the F1-score for Pineal tumour and Choroid plexus tumour appears moderate, that class has only three validation samples, so the estimate is unstable and should be interpreted with caution.

This confirms that the main weakness of the image-features-only approach lies in its handling of minority classes. The model captures useful information, but not enough to provide robust discrimination across all tumor types.

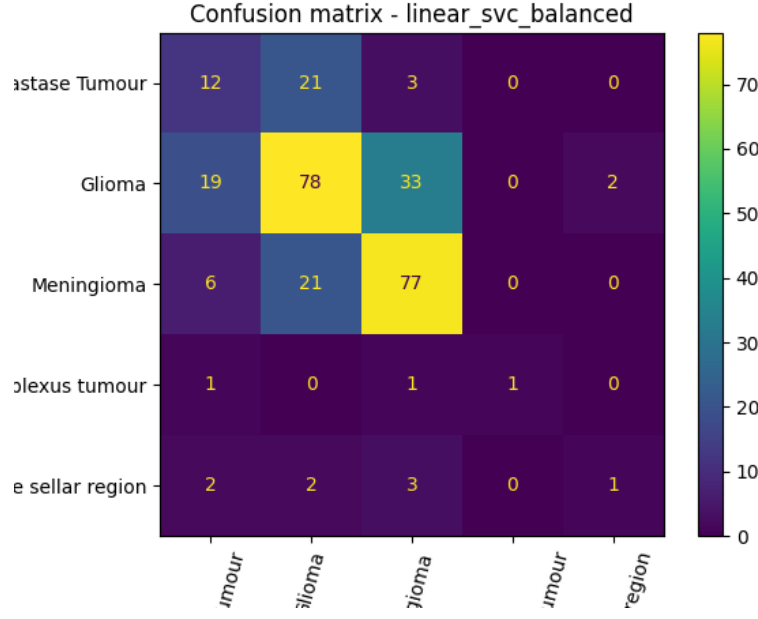


Figure 3: Confusion matrix of the best unimodal model trained on `image_features`. The confusion is concentrated in minority classes, while the dominant classes are recognized more reliably.

3.4 Image Feature Sensitivity Analysis (Ablation Study)

Following the baseline results, we observed that while the combined image feature vector (8,192 dimensions) yielded a Macro F1-score of 0.4617, passing all four MRI modalities simultaneously results in an excessively high-dimensional feature space. To understand the individual predictive power of each MRI modality and to identify potential feature redundancies, we conducted an ablation study using the balanced Linear SVC model. The 8,192-dimensional vector was isolated into four 2,048-dimensional components.

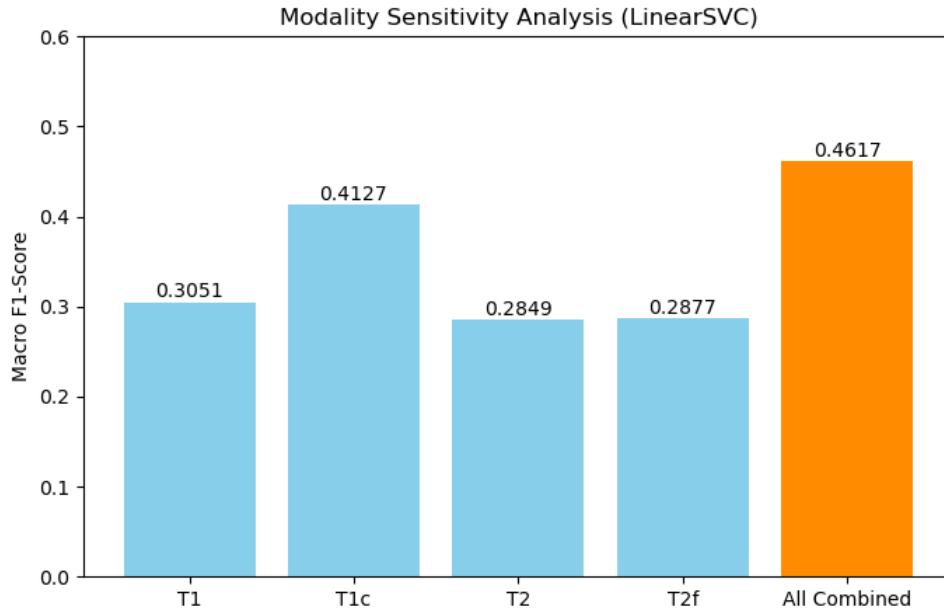


Figure 4: Macro F1-scores across isolated MRI modalities. `ax_t1c` contains the vast majority of the predictive signal.

Configuration	Features Used	Macro F1-Score	% of Full Model Performance
All Combined	T1, T1c, T2, T2f	0.4617	100.0%
ax_t1c	T1-Contrast	0.4127	89.38%
ax_t1	T1-Weighted	0.3051	66.08%
ax_t2f	FLAIR	0.2877	62.31%
ax_t2	T2-Weighted	0.2849	61.71%

Table 3: Ablation study of individual MRI modalities using Linear SVC.

The ablation study reveals the primacy of the `ax_t1c` (T1-Contrast) modality, which alone captures nearly 90% of the predictive power of the combined system. Clinically, this aligns with the fact that contrast enhancement is a primary indicator of blood-brain barrier disruption. The relatively small performance gap (+0.049) between T1c alone and the combined model suggests that the additional 6,144 dimensions from T1, T2, and T2f introduce significant redundancy and mathematical noise.

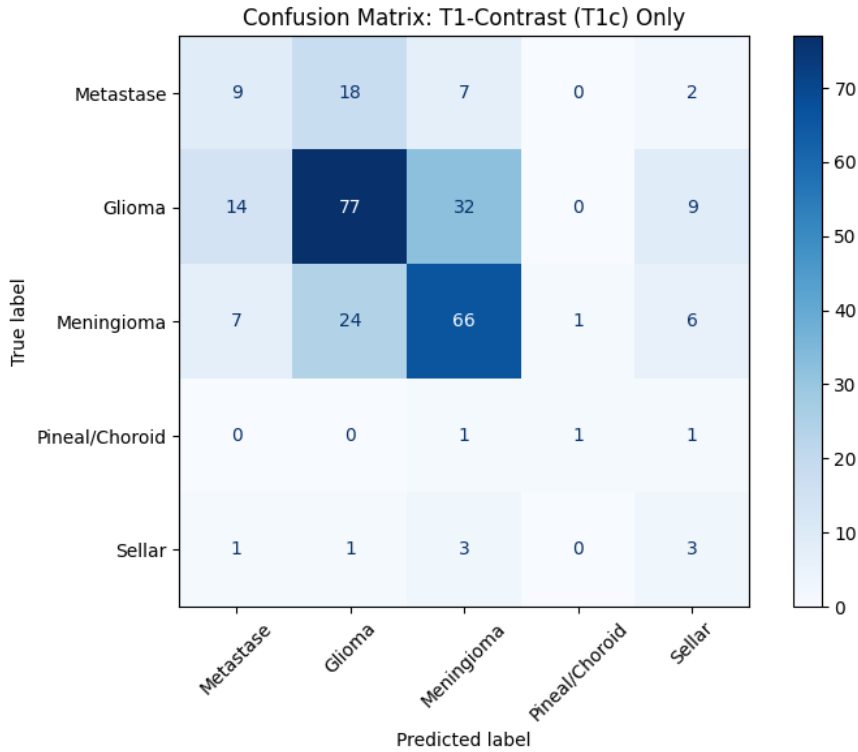


Figure 5: Confusion matrix for the T1c-only model. The error distribution strongly mirrors the combined 8,192-dimensional model, highlighting feature redundancy.

3.5 Dimensionality Reduction via PCA combining with MLP

To mitigate the high-dimensional noise identified in the ablation study, we hypothesized that compressing the feature space could improve the signal-to-noise ratio and prevent overfitting. We applied Principal Component Analysis (PCA) to reduce the 8,196-dimensional image vector down to 256 principal components. These components were then passed into a Multi-Layer Perceptron (MLP). To counteract the severe class imbalance, we utilized a weighted Cross-Entropy loss.

Part 1 (v3): PCA 256 + Weighted Loss + Early Stopping				
	precision	recall	f1-score	support
Brain Metastase Tumour	0.32	0.50	0.39	36
Glioma	0.70	0.51	0.59	132
Meningioma	0.62	0.71	0.66	104
Pineal tumour and Choroid plexus tumour	0.00	0.00	0.00	3
Tumors of the sellar region	0.43	0.38	0.40	8
accuracy			0.57	283
macro avg	0.41	0.42	0.41	283
weighted avg	0.61	0.57	0.58	283

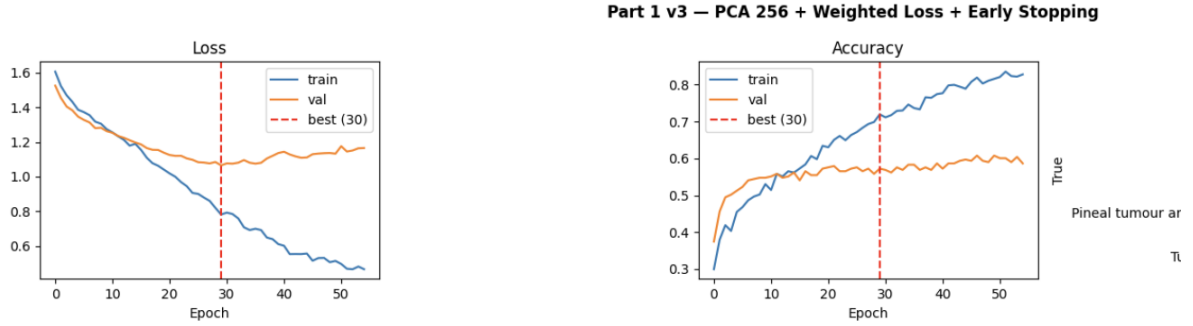


Figure 6: Performance metrics and learning curves for the PCA + MLP model. The divergence between training and validation loss (left) and accuracy (center) confirms persistent overfitting despite dimensionality reduction.

3.5.1 Results and Analysis

Contrary to our hypothesis, the PCA + MLP approach resulted in a performance degradation compared to our linear baseline. The model achieved a **Micro F1-score (Accuracy) of 0.57** and a **Macro F1-score of 0.41** (down from the Linear SVC’s 0.5972 and 0.4617, respectively).

Two key observations explain this drop in performance:

1. **Information Loss in PCA:** Examining the class-wise performance reveals that the MLP completely failed on the minority *Pineal tumour and Choroid plexus tumour* class (F1-score of 0.00, compared to 0.50 in the baseline SVC). PCA compresses data by maximizing variance. Because Gliomas and Meningiomas heavily dominate the dataset, the 256 principal components inherently prioritize their features. The subtle, nuanced variance required to identify the rarest classes was mathematically discarded as “noise” during compression.
2. **Persistent Overfitting:** Despite reducing the feature space from 8,196 to 256 dimensions, the training learning curves indicate persistent overfitting. Training accuracy steadily climbed above 80% while validation accuracy plateaued at 57%, with validation loss increasing after epoch 30.

These results establish a clear limitation of the pre-extracted 1D ResNet embeddings: retaining the full 8,196 dimensions preserves necessary minority class signals but invites noise, while compressing the dimensions via PCA disproportionately destroys minority class information without fully resolving the overfitting of non-linear models.

3.6 Radiomics

Using the same reasoning above, the same independent analysis is performed on radiomics data, the result is as follows.

Knowing that random guessing with largest probability will result in **f1_micro** of 0.48, solely using **f1_micro** is not that significant in prediction.

Model	f1_macro	f1_weighted	f1_micro
random_forest_balanced	0.219987	0.448209	0.522876
extra_trees_balanced	0.192911	0.403909	0.483660
xgb_balanced	0.189461	0.392815	0.450980
logreg_balanced	0.183650	0.255886	0.235294
linear_svc_balanced	0.176985	0.323804	0.294118

Table 4: Radiomics Model performance

3.7 Textual & Clinical Feature Engineering

3.7.1 Clinical and Textual Data Preprocessing

For categorical variables such as gender, tumor anatomical location, T1, enhanced T1 (T1c), T2, and FLAIR sequence signal intensity, we use a label encoder to perform numerical conversion. Age, as a continuous variable, is first filled with missing values using the median, and then normalized to ensure data distribution consistency. All feature scaling operations are fitted only to the training set at each fold of cross-validation, fundamentally avoiding data leakage.

3.7.2 Dual-Pathway Textual Feature Extraction

We designed two complementary feature extraction pathways to transform unstructured radiology reports into modellable quantitative features. The first pathway is a statistical learning method based on TF-IDF vectorization, which converts the report text into numerical features according to word frequency-inverse document frequency, ultimately retaining the top 1000 features, balancing semantic information completeness and computational efficiency. The second pathway is clinical knowledge-driven keyword extraction, building a lightweight rule-matching framework to target and extract 21 imaging features highly relevant to the diagnosis of intracranial tumors, specifically divided into three categories: first, those related to space-occupying effects, including midline shift and brain herniation; second, those related to enhancement patterns, covering ring enhancement and marked enhancement; and third, those related to tumor internal structure, such as necrosis, cystic degeneration, and signal homogeneity/heterogeneity.

3.7.3 Feature Distribution Analysis and Statistical Validation

Frequency Distribution of Extracted Keywords

We first statistically analyzed the occurrence frequency of the 21 imaging feature keywords in the training set (Figure 7). The results showed significant differences in the frequency of different signs: edema (**feat_edema**) occurred most frequently, over 900 times; signal heterogeneity, space-occupying lesion compression, and signal homogeneity all occurred over 200 times; midline shift, dural tail sign, necrosis, and calcification occurred very infrequently, some less than 10 times, providing direct reference for subsequent feature selection.

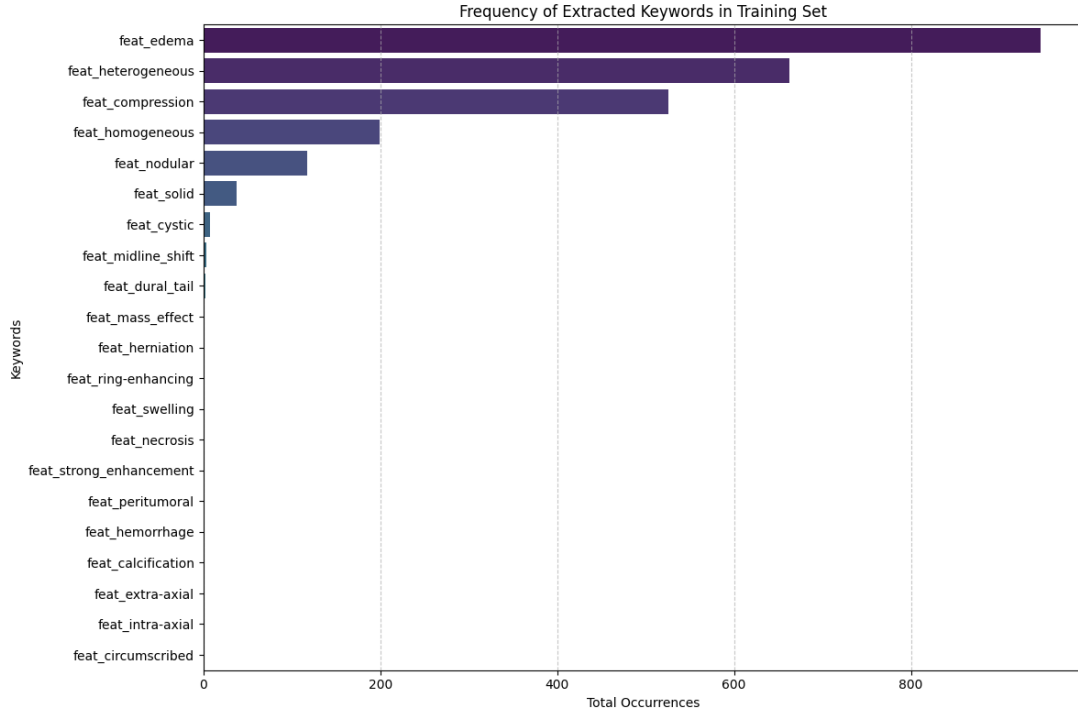


Figure 7: Frequency of Extracted Keywords in Training Set

Differences in Keyword Distribution Across Different Tumor Categories

We statistically analyzed the occurrence proportions of each keyword in five intracranial tumor subtypes (Figure 8), finding significant differences in the distribution of core features across different subtypes: edema was most prevalent in pineal and choroid plexus tumors (87%), followed by brain metastases, sellar region tumors, and gliomas at 78%, 64%, and 57%, respectively, while meningiomas showed only 23%; signal homogeneity was only present in meningiomas (27%), with the remaining subtypes all below 5%; signal heterogeneity was highest in sellar region tumors (54%), with the other four subtypes ranging from 27% to 40%; space-occupying lesion compression was highest in meningiomas (33%), with the remaining subtypes all below 30%. These results confirm that the extracted keyword features can effectively distinguish different tumor subtypes and possess classification predictive value.

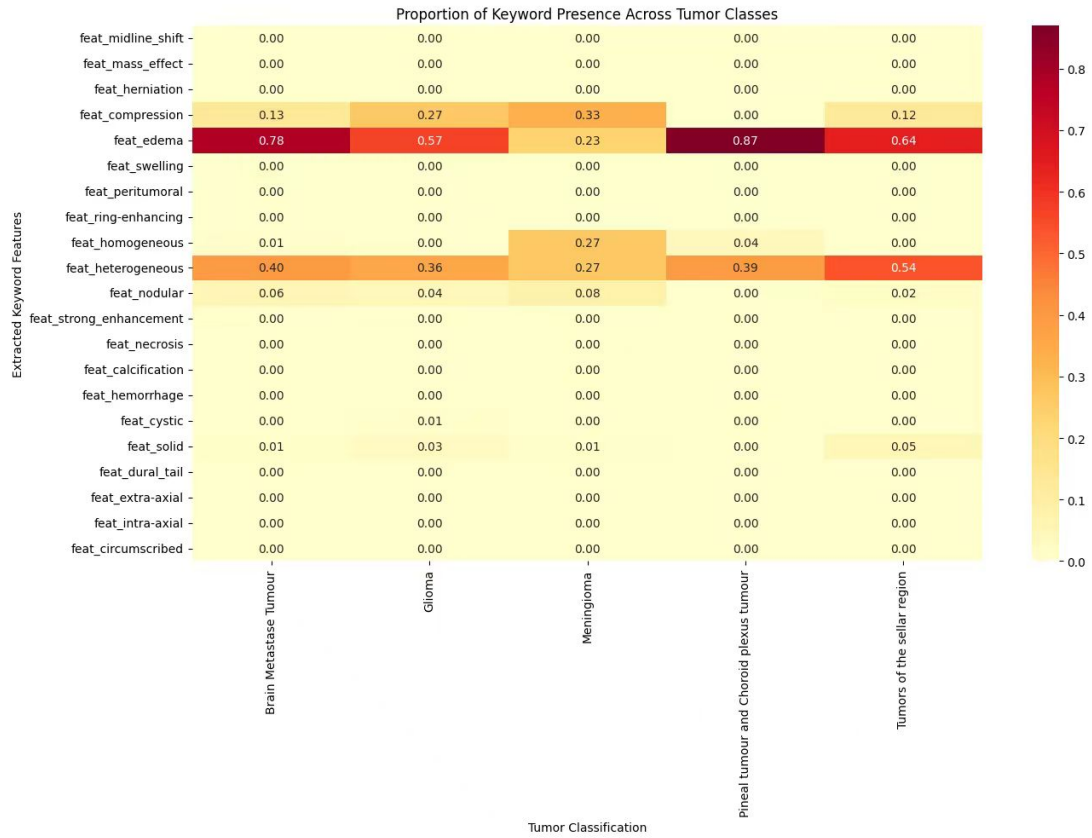


Figure 8: Proportion of Keyword Presence Across Tumor Classes

Statistical Significance Verification of Features

We used the chi-square test to verify the correlation between keyword features and tumor classification labels (Figure 9). Using $p < 0.05$ as the significance level, the results showed that six features – signal homogeneity, edema, space-occupying lesion compression, signal heterogeneity, solid component, and nodularity – were statistically significant. Among these, the chi-square statistics for signal homogeneity and edema were 352.13 and 319.91, respectively, corresponding to p-values of only 6.08×10^{-75} and 5.48×10^{-68} , demonstrating extremely strong predictive value for tumor classification. The p-values for cystic degeneration, dural tail sign, and midline shift were all greater than 0.05, lacking statistical significance, and were subsequently excluded.

	Feature	Chi2_Stat	p_value	Significant (p<0.05)
3	feat_homogeneous	352.131524	6.077790e-75	True
2	feat_edema	319.912872	5.476599e-68	True
1	feat_compression	52.781653	9.467054e-11	True
4	feat_heterogeneous	32.180322	1.757430e-06	True
7	feat_solid	14.855056	5.011477e-03	True
5	feat_nodular	14.823141	5.082449e-03	True
6	feat_cystic	8.051148	8.972245e-02	False
8	feat_dural_tail	3.451283	4.853244e-01	False
0	feat_midline_shift	3.443521	4.865179e-01	False

Figure 9: Chi-square Test Results for Keyword Features

3.7.4 Feature Optimization and Dataset Construction

Based on the above analysis, we performed feature optimization using a two-step method to streamline the feature set and filter noise: First, we used a variance threshold to filter features with no discriminative power, eliminating non-significant features with extremely low frequency and variance less than 0.01; second, using a threshold of 0.85, we removed highly collinear redundant features through the correlation matrix to avoid collinearity interfering with model training and feature importance assessment. The final constructed tabular modality dataset integrates clinical metadata features, statistically validated keyword features, and TF-IDF textual statistical features, providing high-quality input for subsequent model training.

3.7.5 Heterogeneous Tree Ensemble Model for Tabular Modality

For tabular modality data, we constructed a heterogeneous tree ensemble model incorporating LightGBM, Random Forest, and XGBoost. Compared to neural networks, tree models are more robust in capturing nonlinear feature relationships and handling mixed-type structured data and are less prone to overfitting on small sample datasets. The optimized LightGBM single model achieved an F1-score nearly 27% higher than the logistic regression baseline, validating the rationality of the model selection. We optimized Bayesian hyperparameters using the Optuna framework, focusing on adjusting the L1 regularization coefficient `reg_alpha` and the L2 regularization coefficient `reg_lambda` to further mitigate the risk of overfitting on small samples. Hierarchical K-fold cross-validation was used for model training to ensure that the distribution of the five tumor subtypes in each fold of the training set was consistent with the full dataset, guaranteeing the reliability of the cross-validation results. The final model achieved an average Macro F1-score of 0.74 after five-fold cross-validation.

3.7.6 Model Interpretability Analysis

We used SHAP values to quantify the contribution of each feature to the model’s classification decision, calculating the average impact of each feature on the model output based on the mean absolute SHAP value (Figure 10). The results showed that the TF-IDF feature `tfidf_166` had the greatest impact on the model’s decision-making, with a mean absolute SHAP value exceeding 2.0. This was followed by tumor anatomical location encoding features, with a mean absolute SHAP value of 1.2, making it a core clinical feature. Furthermore, T1c sequence signal intensity, patient age, multiple TF-IDF features, and statistically validated signal homogeneity features all significantly contributed to the model’s decision-making. The core driving features identified by the model are highly consistent with the logic of clinical radiological diagnosis: the anatomical location of intracranial tumors is the core basis for clinical qualitative diagnosis, with different subtypes having characteristic predilection sites; the MRI signal characteristics, enhancement patterns, and peritumoral edema of the tumor are key indicators for clinically differentiating between benign and malignant tumors and determining pathological subtypes. This result confirms that the model’s decision-making logic has sufficient clinical basis, enhancing the model’s academic rigor and clinical credibility.

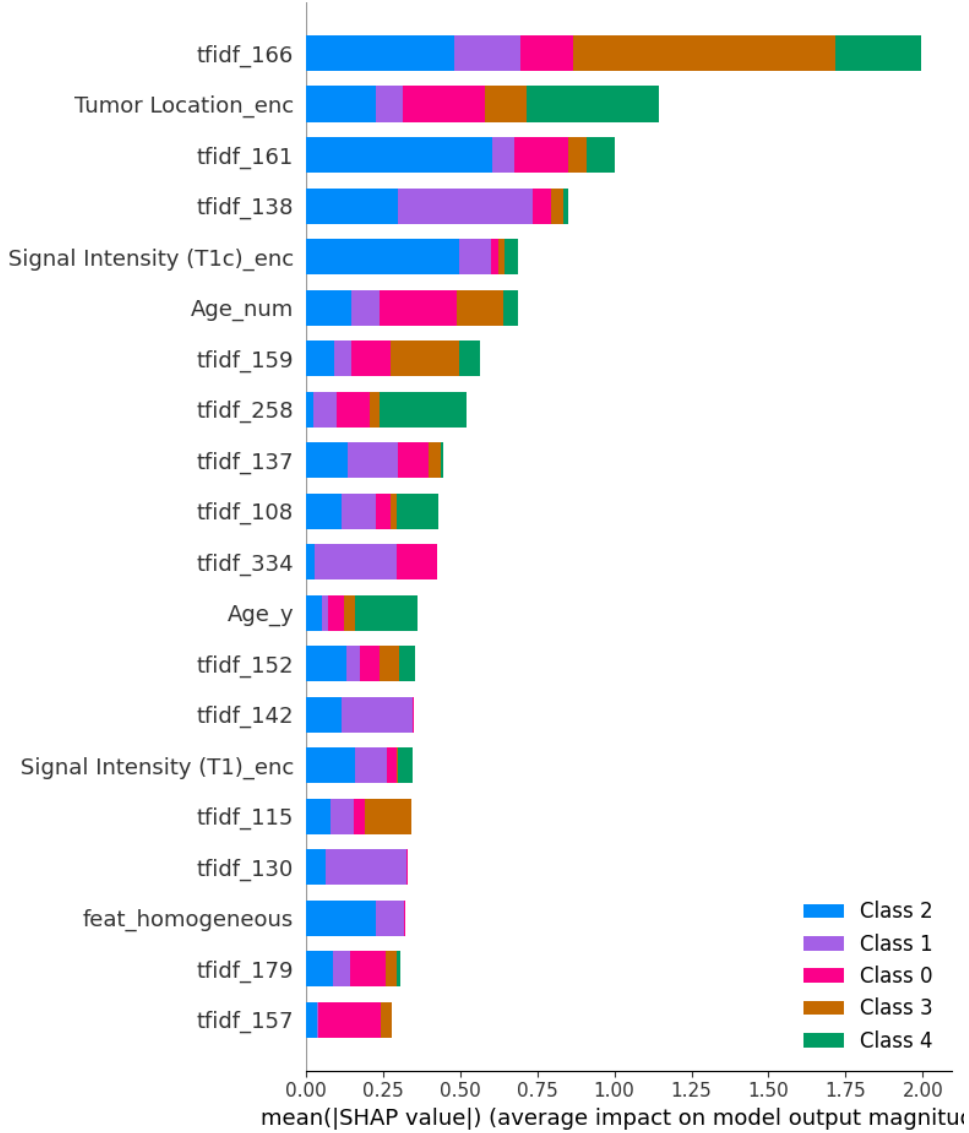


Figure 10: Mean Absolute SHAP Values of Top Features for the Tabular Modality Model

3.7.7 Output Design for Multimodal Fusion

For high-dimensional TF-IDF features, we employ PCA for dimensionality reduction, mitigating the curse of dimensionality while preserving core semantic information, thus preparing for subsequent multimodal feature-level fusion. The tabular modality model ultimately outputs soft classification probabilities for 5 tumor subtypes, rather than hard classification labels. The model achieves an average classification confidence of 71.4 on the test set, providing a quantitative basis for subsequent decision-level weight fusion with the image modality model. These interpretable features, along with deep image features, are input into the final voting classifier to achieve multimodal collaborative decision-making and improve the overall performance of tumor classification.

3.8 Interpretation

Several conclusions can be drawn from this study of the `image_features` modality.

First, deep MRI embeddings extracted by ResNet do contain meaningful predictive information, since the best image-only model reaches a non-trivial Micro F1-score of 0.5972. Furthermore, class imbalance has a strong effect on performance.

More broadly, this analysis shows both the usefulness and the limitations of the deep image features. On the one hand, they provide a solid unimodal baseline and are especially informative for common

classes. On the other hand, they are insufficient on their own to achieve strong performance across all tumor categories. This strongly motivates multimodal fusion with radiomics, clinical variables, and radiology report text.

4 Soft Voting and Naive Bayes Ensemble

4.1 Motivation

The results from the separate training stage showed that each MRI modality contains useful but incomplete information for tumor classification. However, due to missing modalities and class imbalance, no single model is sufficient to achieve robust performance across all classes.

To address this limitation, we explored ensemble methods that combine predictions from multiple modality-specific models. The intuition is that different modalities capture complementary aspects of the tumor, and aggregating their predictions can lead to more reliable and stable classification.

4.2 Per-modality MLP Models

Instead of training a single model on concatenated features, we trained one Multi-Layer Perceptron (MLP) for each MRI modality (`ax_t1`, `ax_t1c`, `ax_t2`, `ax_t2f`). Each model was trained only on the subset of patients for which the corresponding modality was available.

Each MLP takes a 2048-dimensional deep feature vector as input and consists of multiple fully connected layers with batch normalization, ReLU activation, and dropout regularization. The final layer outputs class probabilities through a softmax function.

This design allows each model to specialize in learning modality-specific patterns, while naturally handling missing data by excluding unavailable modalities during both training and inference.

4.3 Ensemble Strategies

At inference time, each available modality-specific model produces a probability vector over the five tumor classes. These predictions are then combined using two different ensemble strategies.

4.3.1 Soft Voting

In the Soft Voting approach, the predicted probability vectors from all available modalities are averaged:

$$\hat{p} = \frac{1}{M} \sum_{i=1}^M p_i$$

where M is the number of available modalities for a given patient and p_i is the probability vector predicted by the i -th model.

The final prediction is obtained by taking the class with the highest averaged probability. This method assumes that all modalities contribute equally and provides a smooth aggregation of model outputs.

4.3.2 Naive Bayes Ensemble

In the Naive Bayes approach, the probability vectors are combined multiplicatively under an independence assumption:

$$\hat{p} \propto \prod_{i=1}^M p_i$$

To ensure numerical stability, this is implemented in log-space:

$$\log \hat{p} = \sum_{i=1}^M \log p_i$$

The final prediction is obtained by selecting the class with the highest combined log-probability. This method emphasizes agreement across modalities and tends to produce more confident predictions when multiple models support the same class.

4.4 Handling Missing Modalities

A key advantage of this framework is its flexibility in handling missing data. For each patient, only the modalities that are available are used in the ensemble. If a modality is missing, its corresponding model simply does not contribute to the final prediction.

This avoids the need for imputation or zero-padding at the model level and ensures that predictions are based only on observed data.

4.5 Extension with Radiomics Features

In a further extension, we incorporated radiomics features as an additional modality. A separate MLP was trained on a 20-dimensional radiomics feature vector derived from four MRI modalities.

During inference, this radiomics model acts as an additional voter alongside the four image-based models. As with the image modalities, the radiomics prediction is included only when the corresponding data is available.

This results in a five-model ensemble that integrates both deep image features and handcrafted radiomics features.

4.6 Discussion

The ensemble framework provides several advantages over unimodal models. First, it leverages complementary information from different modalities, which is particularly important for difficult or ambiguous cases. Second, it improves robustness to missing data by dynamically adapting to the available inputs.

Comparing the two strategies, Soft Voting tends to produce more stable predictions by averaging across models, while the Naive Bayes approach can yield sharper decisions by reinforcing agreement between modalities. However, the independence assumption in Naive Bayes may not always hold in practice, as different MRI modalities are often correlated.

Overall, these ensemble methods represent a simple yet effective step toward multimodal integration, bridging the gap between separate training and more advanced fusion techniques.

5 Conclusion

In this project, we developed statistical machine learning models to tackle the challenging five-class presurgical brain tumor classification problem using multimodal data[cite: 8, 9]. Our initial unimodal analysis demonstrated that while deep MRI embeddings contain meaningful predictive signals, they are insufficient on their own to achieve strong performance across all categories due to severe class imbalance[cite: 660, 661, 665]. Specifically, our ablation study revealed that the T1-contrast (T1c) modality is the most critical, capturing nearly 90% of the predictive power of the combined image features[cite: 220]. Attempts to mitigate high dimensionality using PCA and MLP resulted in information loss for minority classes and persistent overfitting, highlighting the limitations of simply compressing these representations[cite: 357, 360, 362].

Conversely, our feature engineering on clinical and textual data proved highly effective. By integrating TF-IDF vectorization with clinical knowledge-driven keyword extraction (such as edema and signal homogeneity), our heterogeneous tree ensemble achieved a robust Macro F1-score of 0.74[cite: 377, 378, 610, 616]. Furthermore, SHAP interpretability analysis validated this approach, confirming that the model’s decision-making heavily relies on critical features like tumor location and T1c signal intensity, which perfectly aligns with clinical radiological diagnosis logic[cite: 618, 621, 624].

Finally, to effectively integrate these diverse data sources and naturally handle the issue of missing patient modalities, we implemented Soft Voting and Naive Bayes ensemble frameworks[cite: 679, 682, 688, 699]. By aggregating probability predictions from modality-specific MLPs and radiomics models, our ensemble strategies successfully leverage complementary multimodal information[cite: 706, 708, 709]. Overall, this multimodal fusion approach bridges the gap between isolated feature sets, offering a flexible, interpretable, and robust solution for complex brain tumor classification[cite: 712].

References

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